

Basant Bhushan

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SKILLS SUMMARY

Languages:	C++, JavaScript, TypeScript, SQL, Python
Frameworks & Libraries:	React, Next.js, Node.js, Express.js, Flask, Prisma
Databases:	MongoDB, PostgreSQL
AI / APIs:	Google Gemini API, YouTube Data API, Google OAuth 2.0, REST APIs
Cloud / Tools:	Git, GitHub, Render, Vercel, PyGithub
Core Subjects:	Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks

PROJECTS

Tunelt – Algorithmic Music Flow Sequencer & Energy Optimizer | [GitHub](#) | [Live](#) May 2026 – Aug 2026

- Architected 4 algorithmic sequencing engines using Camelot harmonic matching and regression-based energy optimization, tested and refined on music-analysis datasets including MusAV and MTG-Jamendo, achieving 157K+ tracks/sec throughput and 99.7% smoothness on a 500-track benchmark.
- Engineered an Express.js/TypeScript REST API integrating YouTube Data API and Google Gemini, with 12-track batching, **Bottleneck rate limiting** and PostgreSQL/Prisma caching for efficient track analysis.
- Developed a Next.js 16 frontend with interactive energy-trajectory visualizations, in-browser audio previews, and Google OAuth 2.0 playlist synchronization.

Tech Stack: TypeScript, Next.js, Express.js, PostgreSQL, Prisma, Google Gemini API, YouTube Data API, Google OAuth 2.0

OnSight – Automated Examination & Assessment Platform | [GitHub](#) Mar 2026 – Apr 2026

- Engineered an automated PDF-to-MCQ ingestion pipeline using Gemini multimodal APIs, with JSON schema validation, input sanitization, and fallback handling for transient extraction failures.
- Developed a Node.js/MongoDB grading engine that evaluates submissions, enforces exam time windows, computes question-level performance analytics, and supports post-exam grace-mark recalculations.
- Architected a secure backend with hierarchical RBAC, JWT authentication and refresh-token rotation, endpoint rate limiting, and persistent audit logging for unauthorized access attempts.

Tech Stack: Node.js, Express.js, MongoDB, Mongoose, Gemini Multimodal API, RBAC, JWT, REST APIs, JavaScript

BobAI – Natural-Language Full-Stack Application Generator | [GitHub](#) Feb 2026 – May 2026

- Architected a Flask REST API and modular scaffolding engine that converts natural-language specifications into multi-tier applications across vanilla JS, React, and Node/Express with generated SQL schemas.
- Engineered a multi-file code extraction pipeline using Gemini with regex fallbacks, integrating Pexels APIs to inject contextual assets and design tokens into generated source files.
- Automated repository provisioning and GitHub deployment using PyGithub, with subprocess-based integration tests validating API contracts, generated-file integrity, and deployment failure handling.

Tech Stack: Python, Flask, Google Gemini API, PyGithub, GitHub API, REST APIs, React, Node.js, SQL

TRAININGS

UI/UX Design and Prototyping for Web and Mobile Applications | [Certificate](#) Jun 2026 – Jul 2026

- Learned core UI/UX principles — wireframing, prototyping, and usability testing — and applied them to design responsive interfaces for web and mobile applications.

Centre for Professional Enhancement, Lovely Professional University — Grade: A

CERTIFICATIONS

Cloud Fundamentals with Proctored Exam | [Certificate](#) Mar 2025

Web Development (Live MOOC) | [Certificate](#) Jan 2025

EDUCATION

Lovely Professional University Punjab, India
Bachelor of Technology in Computer Science and Engineering — CGPA: 6.75 Aug 2024 – Jun 2028

Milton Public School Agra, India
Senior Secondary (Class XII) — 67.7% 2024